

EEE/CPE 64 — Lab Report 7

2-BIT COMPARATOR

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INTRODUCTION

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This lab introduces the design, simulation, and hardware implementation of a 2-bit magnitude comparator using Multisim and an FPGA board. I first construct the truth table and a gate-level schematic, then verify the comparator in Multisim with switches as inputs and LEDs showing the binary outputs for the required test cases. Finally, I load the provided Verilog into Quartus, map inputs to FPGA switches and outputs to LEDs, and confirm the same scenarios on hardware, demonstrating how the comparator indicates $A > B$, $A = B$, or $A < B$.

PRELIMINARY CALCULATIONS

2.1 GATE-LEVEL SCHEMATICS AND CALCULATIONS

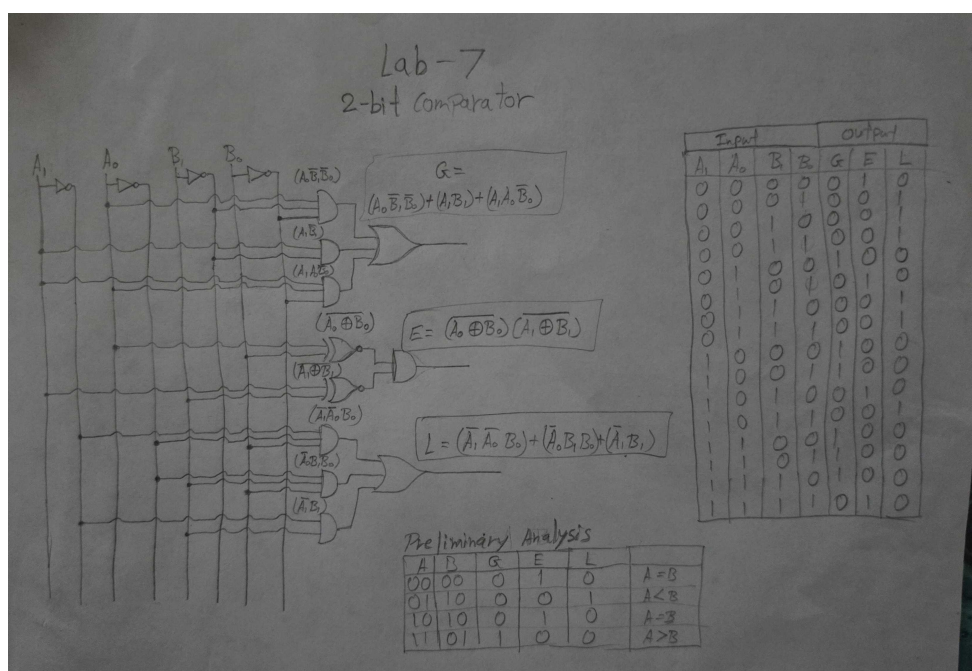


Figure 1: Gate level Schematics

A 2-bit comparator compares two 2-bit numbers by looking at the left (most-significant) bits first. If those are different, the one with 1 is larger and the result is decided. If they're the same, it then looks at the right (least-significant) bits to decide larger or smaller; if both bits match, the numbers are equal. This is the logic used to predict each case in the calculations.

SIMULATION RESULTS

3.1 NI MULTISIM: SCHEMATICS

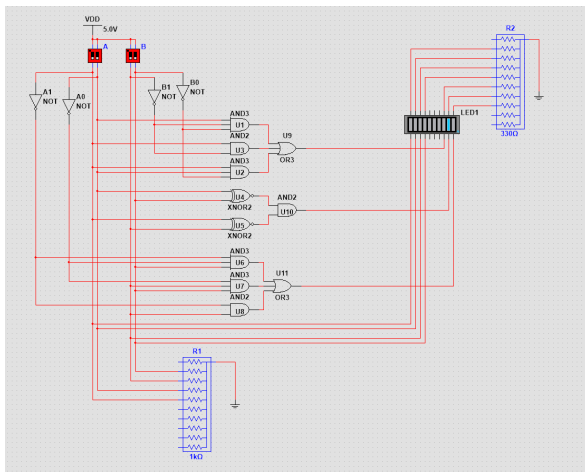
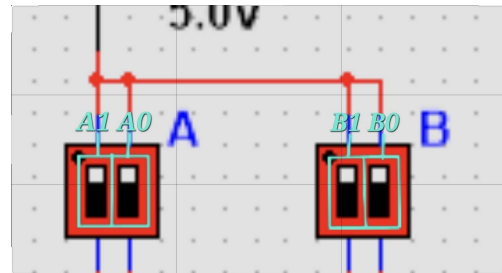
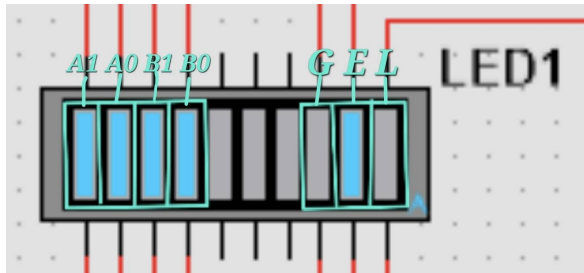


Image 5

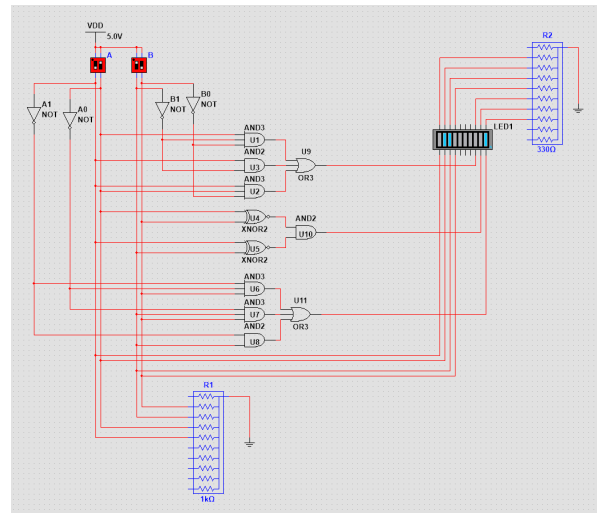


Image 6

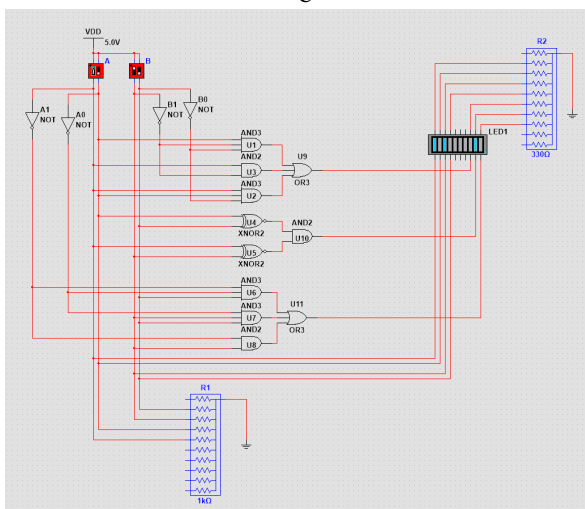


Image 7

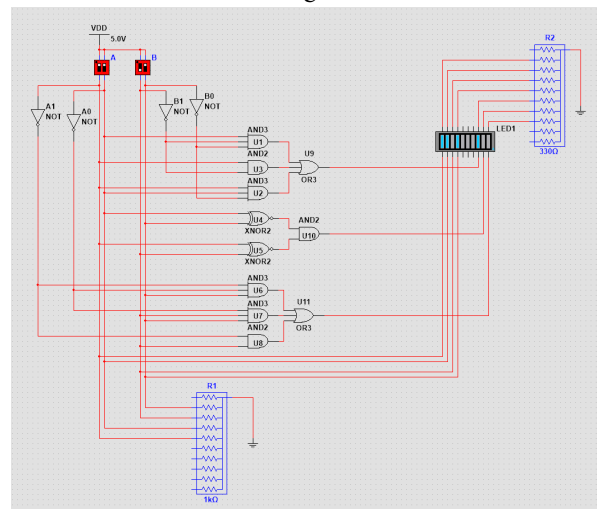


Image 8

Figure 3: 2-bit Comparator

FPGA RESULTS

4.1 FPGA DATA

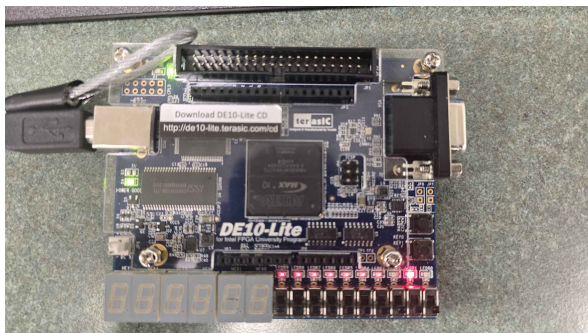


Image 5

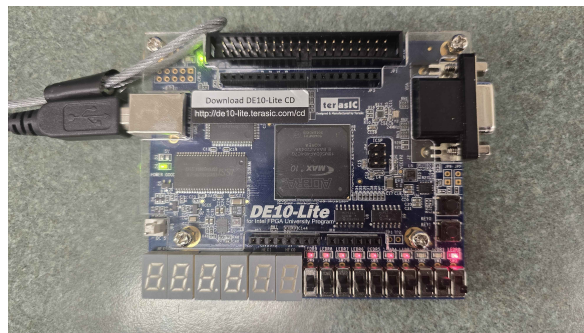


Image 6



Image 7

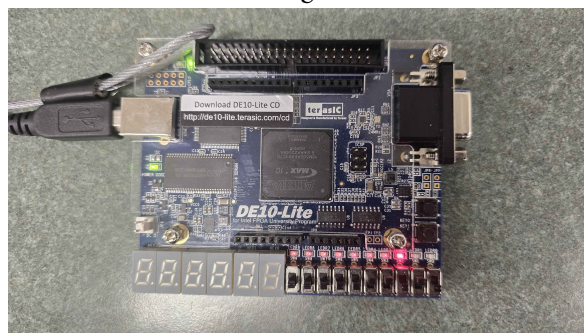


Image 8

Figure 4: 2-bit Comparator FPGA

CONCLUSION

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We validated the 2-bit magnitude comparator in three stages: preliminary calculations, NI Multisim simulation, and FPGA implementation in Quartus. For all test cases (00, 00), (01, 10), (10, 10), (11, 01), the predicted relations $A > B$, $A = B$, and $A < B$ matched exactly across every stage. This agreement confirms the correctness of the derived logic and the gate-level design.

A	B	Preliminary Analysis	Simulation Results	FPGA Results
00	00	$A = B$	$A = B$	$A = B$
01	10	$A < B$	$A < B$	$A < B$
10	10	$A = B$	$A = B$	$A = B$
11	01	$A > B$	$A > B$	$A > B$

Table 1: 2-bit comparator results confirmed across preliminary work, Multisim, and FPGA.